

AI-Based Adaptive Battlefield Awareness & Decision Support System (ABADS)

Project Proposal / Mini Thesis

1. Introduction

This project focuses on developing an intelligent battlefield support system using a UGV, sensors, and AI to detect threats, map risk zones, and suggest safe navigation paths.

2. Objective

To design a system that reduces soldier risk by integrating detection, mapping, and decision-making into a unified platform.

3. Problem Statement

Manual mine detection is risky and inefficient. There is a need for a system that can autonomously detect threats and assist in decision-making.

4. Proposed Solution

The ABADS system uses a UGV with sensors to detect mines, process data using AI, generate a risk map, and suggest safe paths.

5. System Components

Hardware: UGV chassis, sensors, Arduino, Raspberry Pi. Software: Python, OpenCV, SQL.

6. Methodology

Step 1: Robot navigation. Step 2: Sensor detection. Step 3: AI processing. Step 4: Risk mapping. Step 5: Path planning.

7. Applications

Military minefield detection, disaster rescue, hazardous area navigation.

8. Advantages

Reduces human risk, improves decision-making, integrates multiple technologies.

9. Limitations

Prototype limitations, basic AI, not military-grade system.

10. Conclusion

ABADS demonstrates an integrated approach to battlefield awareness and decision support, aligning with modern defence needs.